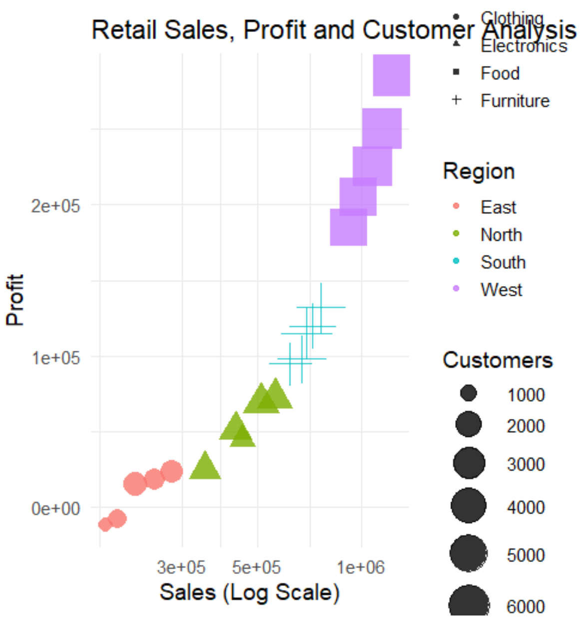
Question 1

A multinational retail company has collected annual sales information from its branches across different regions. Using the given dataset, design an effective visualization that simultaneously represents Sales, Profit, Number of Customers, Product Category, and Region. Select suitable aesthetic mappings and apply an appropriate scale transformation wherever necessary. Justify every visualization design choice.

```
# Load package
library(ggplot2)

# Dataset
retail <- data.frame(
Branch=c("B1","B2","B3","B4","B5","B6","B7","B8","B9","B10",
"B11","B12","B13","B14","B15","B16","B17","B18","B19","B20"),
Region=c("North","South","East","West","North","South","East","West",
"North","South","East","West","North","South","East","West",
"North","South","East","West"),
Category=c("Electronics","Furniture","Clothing","Food","Electronics",
"Furniture","Clothing","Food","Electronics","Furniture",
"Clothing","Food","Electronics","Furniture","Clothing",
"Food","Electronics","Furniture","Clothing","Food"),
Sales=c(450000,620000,180000,920000,350000,720000,250000,1080000,
560000,670000,195000,980000,430000,760000,280000,1150000,
510000,690000,220000,1250000),
Profit=c(45000,95000,-12000,185000,25000,120000,18000,225000,
72000,98000,-8000,205000,51000,132000,24000,250000,
69000,115000,15000,285000),
Customers=c(1200,2500,980,4200,1800,3000,1250,5200,
2100,3300,1100,4700,1750,3600,1450,5600,
2200,3900,1600,6100)
)
```

```
# Bubble Chart
ggplot(retail, aes(x = Sales,
                   y = Profit,
                   size = Customers,
                   color = Region,
                   shape = Category)) +
  geom_point(alpha = 0.8) +
  scale_x_log10() +
  scale_size(range = c(4, 12)) +
  labs(
    title = "Retail Sales, Profit and Customer Analysis",
    x = "Sales (Log Scale)",
    y = "Profit",
    size = "Customers",
    color = "Region",
    shape = "Product Category"
  ) +
  theme_minimal(base_size = 14)
```



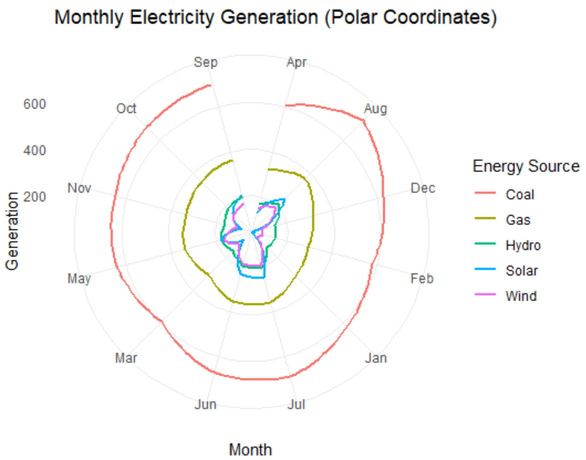
Question 2

The following dataset contains monthly electricity generation from different energy sources. Develop visualizations using Cartesian Coordinates and Polar Coordinates. Compare both visualizations and determine which coordinate system provides better analytical insight. Explain your design decisions.

```
# Load package
library(ggplot2)
library(tidyr)

# Dataset
energy <- data.frame(
Month=month.abb,
Coal=c(620,580,590,605,640,670,690,720,700,680,650,630),
Gas=c(310,295,305,320,340,355,360,370,365,350,335,320),
Solar=c(45,60,90,130,180,230,250,245,210,160,90,50),
Wind=c(80,95,120,150,170,185,195,190,175,150,110,90),
Hydro=c(140,150,160,170,180,195,205,210,200,185,165,150)
)
# Convert to long format
energy_long <- pivot_longer(
  energy,
  cols = -Month,
  names_to = "Source",
  values_to = "Generation"
)
# -------------------------
# Cartesian Coordinate Plot
# -------------------------
ggplot(energy_long,
       aes(x = Month,
           y = Generation,
           color = Source,
```

```
          group = Source)) +
  geom_line(size = 1.2) +
  geom_point(size = 2) +
  labs(
    title = "Monthly Electricity Generation (Cartesian Coordinates)",
    x = "Month",
    y = "Electricity Generation (MW)",
    color = "Energy Source"
  ) +
  theme_minimal()
# -------------------------
# Polar Coordinate Plot
# -------------------------
ggplot(energy_long,
       aes(x = Month,
          y = Generation,
          color = Source,
          group = Source)) +
  geom_line(size = 1) +
  coord_polar() +
  labs(
    title = "Monthly Electricity Generation (Polar Coordinates)",
    color = "Energy Source"
  ) +
  theme_minimal()
```

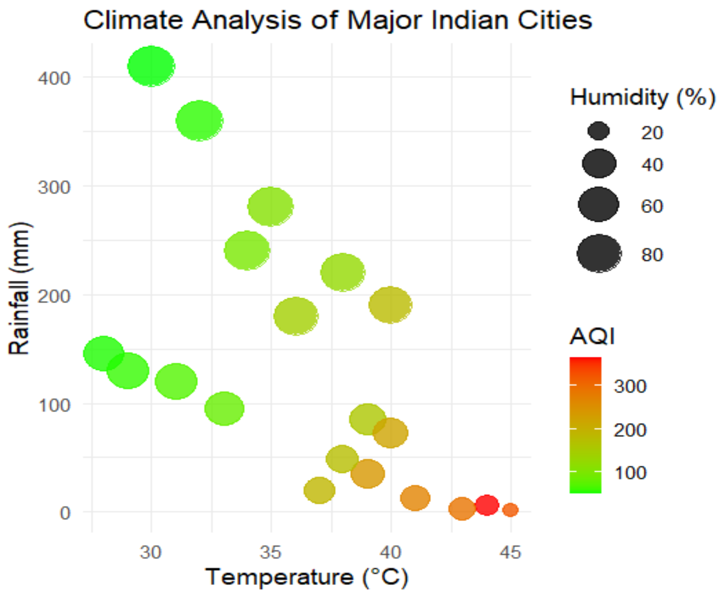


Question 3

An environmental monitoring agency has collected climate data from major Indian cities. Design a visualization that effectively represents Temperature, Air Quality Index (AQI), Rainfall, and Humidity using appropriate colour scales. Select a suitable colour palette and justify its effectiveness for accurate interpretation and accessibility.

```
# Load package
library(ggplot2)
# Dataset
climate <- data.frame(
City=c("Delhi","Mumbai","Chennai","Kolkata","Hyderabad",
"Bangalore","Pune","Ahmedabad","Jaipur","Lucknow",
"Bhopal","Patna","Goa","Kochi","Nagpur","Indore",
"Surat","Visakhapatnam","Coimbatore","Mysuru"),
Temperature=c(44,35,38,40,39,31,33,43,45,39,
38,40,32,30,41,37,36,34,29,28),
Humidity=c(22,84,78,72,48,66,58,26,18,42,
36,45,89,91,30,34,76,82,68,64),
Rainfall=c(6,280,220,190,85,120,95,2,1,35,
48,72,360,410,12,20,180,240,130,145),
AQI=c(365,110,125,180,160,72,84,290,320,240,
175,220,58,52,270,190,145,98,60,55)
)
# Bubble Chart
ggplot(climate,
       aes(x = Temperature,
           y = Rainfall,
           size = Humidity,
           color = AQI)) +
  geom_point(alpha = 0.8) +
  scale_color_gradient(
    low = "green",
```

```
  high = "red"
 ) +
 scale_size(range = c(4,12)) +
 labs(
  title = "Climate Analysis of Major Indian Cities",
  x = "Temperature (°C)",
  y = "Rainfall (mm)",
  size = "Humidity (%)",
  color = "AQI"
 ) +
 theme_minimal(base_size = 14)
```



Question 4

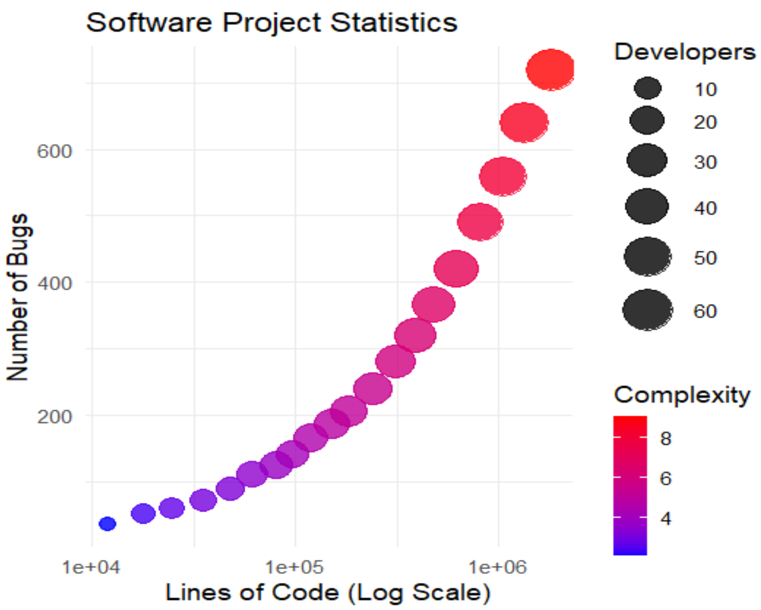
A software company has recorded project statistics for twenty software development teams. Design a visualization that requires the use of a nonlinear axis transformation to represent highly skewed data. Justify the selected transformation and explain how it improves interpretation over a linear scale.

```
# Load package
library(ggplot2)
# Dataset
software <- data.frame(
```

```
Team=paste("T",1:20,sep=""),
LinesOfCode=c(12000,18000,25000,35000,48000,
62000,81000,96000,120000,150000,
185000,240000,310000,390000,480000,
620000,810000,1050000,1350000,1800000),
Bugs=c(35,52,60,72,88,
110,125,142,165,188,
205,240,280,320,365,
420,490,560,640,720),
Developers=c(5,7,9,10,12,
14,16,18,20,22,
25,28,30,34,38,
42,46,50,55,60),
Complexity=c(2.1,2.4,2.7,3.0,3.2,
3.5,3.8,4.1,4.5,4.8,
5.1,5.4,5.8,6.2,6.6,
7.1,7.6,8.0,8.5,9.0)
)

# Bubble Chart with Log Scale
ggplot(software,
       aes(x = LinesOfCode,
           y = Bugs,
           size = Developers,
           color = Complexity)) +
  geom_point(alpha = 0.8) +
  scale_x_log10() +
  scale_color_gradient(low = "blue", high = "red") +
  scale_size(range = c(4,12)) +
  labs(
    title = "Software Project Statistics",
    x = "Lines of Code (Log Scale)",
```

```
   y = "Number of Bugs",
   size = "Developers",
   color = "Complexity"
 ) +
 theme_minimal(base_size = 14)
```



Question 5

A financial analytics company has collected performance data for twenty publicly listed companies. Develop an integrated visualization that effectively communicates Revenue, Profit Margin, Market Capitalization, Employee Strength, and Industry Sector. Use appropriate aesthetic mappings, coordinate systems, scales, and colour palettes to maximise interpretability while minimising perceptual bias.

```
# Load package
library(ggplot2)
# Dataset
companies <- data.frame(
Company=paste("C",1:20,sep=""),
Sector=c("IT","Banking","Healthcare","Retail","Energy",
"IT","Banking","Healthcare","Retail","Energy",
"IT","Banking","Healthcare","Retail","Energy",
"IT","Banking","Healthcare","Retail","Energy"),
```

```
Revenue=c(420,580,720,850,980,
1150,1320,1480,1650,1820,
2050,2280,2550,2840,3150,
3480,3860,4250,4680,5200),
ProfitMargin=c(8.2,15.6,11.4,9.3,18.5,
20.4,17.8,14.2,10.5,22.1,
24.5,19.8,16.7,13.4,26.2,
28.1,21.5,18.3,15.2,30.4),
Employees=c(1500,3200,2100,4200,2800,
5100,6800,3900,7200,4500,
8100,9500,6200,10300,7200,
11800,13200,8600,14500,16000),
MarketCap=c(8500,12000,9800,7600,14500,
18200,21500,16800,13200,24500,
28500,32500,24800,19200,38200,
42500,46800,35600,29800,52000)
)

# Bubble Chart
ggplot(companies,
       aes(x = Revenue,
           y = ProfitMargin,
           size = MarketCap,
           color = Employees,
           shape = Sector)) +
  geom_point(alpha = 0.8) +
  scale_size(range = c(4, 12)) +
  scale_color_gradient(low = "lightblue", high = "darkblue") +
  labs(
    title = "Financial Performance of Publicly Listed Companies",
    x = "Revenue (Million)",
    y = "Profit Margin (%)",
```

```
  size = "Market Capitalization",
  color = "Employees",
  shape = "Industry Sector"
) +
theme_minimal(base_size = 14)
```